

Integraltransformation Block 4

1) b) $f_n = \sin(\omega_0 n)$

Tabelle 19:

$$\rightarrow X(z) = \frac{z \cdot \sin(\omega_0)}{z^2 - 2z \cos(\omega_0) + 1}$$

c) $f_n = \{1, -1, 1, -1, \dots\}$

$$f_n \triangleq (-1)^n \rightarrow \text{Tabelle 9: } (a^k \cdot \varepsilon[k]) \Rightarrow \frac{z}{z-a}$$

$$X(z) = \frac{z}{z-(-1)} = \frac{z}{z+1}$$

2) b) $f_n = e^{an} \sin(\omega_0 n + \varphi)$

Additionssatz: $\sin(\omega_0 n + \varphi) = \sin(\omega_0 n) \cdot \cos(\varphi) + \cos(\omega_0 n) \cdot \sin(\varphi)$

$$f_n = e^{an} \cdot \sin(\omega_0 n) \cdot \cos(\varphi) + e^{an} \cdot \cos(\omega_0 n) \cdot \sin(\varphi)$$

aus Tabelle wird 20 + 21 verwendet (a entspricht dabei aber e^a)

$$a^n \sin(\omega_0 n) \rightarrow \frac{z a \sin(\omega_0)}{z^2 - 2za \cos(\omega_0) + a^2}$$

$$a^n \cos(\omega_0 n) \rightarrow \frac{z \cdot [z - a \cos(\omega_0)]}{z^2 - 2za \cos(\omega_0) + a^2}$$

$$X(z) =$$

$$X(z) = \frac{z e^a \sin(\omega_0) \cos(\varphi)}{z^2 - 2z e^a \cos(\omega_0) + e^{2a}} + \frac{z \cdot (z - e^a \cos(\omega_0)) \sin(\varphi)}{z^2 - 2z e^a \cos(\omega_0) + e^{2a}}$$

$$X(z) = \frac{z e^a \sin(\omega_0) \cos \varphi + z \cdot (z - e^a \cos(\omega_0)) \sin(\varphi)}{z^2 - 2z e^a \cos(\omega_0) + e^{2a}}$$

$$X(z) = \frac{z^2 \sin(\varphi) + z e^a \sin(\omega_0 - \varphi)}{z^2 - 2z e^a \cos(\omega_0) + e^{2a}}$$

2) c) $f_n = n^2 \sigma_n$

Tabelle: 6 $k^2 \cdot \varepsilon[k] \rightarrow \frac{z(z+1)}{(z-1)^3}$

$$X(z) = \frac{z \cdot (z+1)}{(z-1)^3} \stackrel{!}{=} \underline{\underline{z \cdot \frac{(z+1)}{(z-1)^3}}}$$

3) b) $X(z) = \frac{z}{1+z^2}$

$$1+z^2 = (z-j) \cdot (z+j)$$

$$\frac{X(z)}{z} = \frac{1}{(z-j) \cdot (z+j)} \rightarrow \frac{A}{z-j} + \frac{B}{z+j}$$

$$1 = A \cdot (z+j) + B \cdot (z-j)$$

$$1 = (A+B) \cdot z + j \cdot (A-B)$$

$$\begin{aligned} A+B &= 0 \\ A-B &= -j \end{aligned} \rightarrow \begin{aligned} A &= -\frac{j}{2} \\ B &= \frac{j}{2} \end{aligned}$$

$$X(z) = -\frac{j}{2} \cdot \frac{z}{z-j} + \frac{j}{2} \cdot \frac{z}{z+j}$$

Tabelle 9: $\frac{z}{z-a} \rightarrow a^k \cdot \varepsilon[k]$

$$f_n = -\frac{j}{2} \cdot j^k \cdot \varepsilon[k] + \frac{j}{2} \cdot (-j)^k \cdot \varepsilon[k]$$

$$f_n = \frac{j}{2} \cdot \left[-e^{jk\pi/2} + e^{-jk\pi/2} \right] \cdot \varepsilon[k]$$

$$f_n = \frac{j}{2} \cdot \left[-2 \cdot j \sin\left(\frac{k\pi}{2}\right) \right] \cdot \varepsilon[k]$$

$$\underline{\underline{f_n = \sin\left(\frac{k\pi}{2}\right) \cdot \varepsilon[k]}}$$

$$j = e^{j\pi/2}$$

$$-j = e^{-j\pi/2}$$

3) c) $X(z) = \frac{2z^2 + 1}{(z+2) \cdot (z-1)}$

$$\frac{X(z)}{z} = \frac{2z^2 + 1}{z \cdot (z+2) \cdot (z-1)} \stackrel{^}{=} \frac{A}{z} + \frac{B}{z+2} + \frac{C}{z-1}$$

$$2z^2 + 1 = A \cdot (z+2) \cdot (z-1) + Bz(z-1) + Cz(z+2)$$

- $A(z^2 + z - 2)$

- $B(z^2 - z)$

- $C(z^2 + 2z)$

$$2z^2 + 1 = (A+B+C)z^2 + (A-B+2C)z + (-2A)$$

$$A+B+C=2$$

$$A-B+C=0$$

$$-2A = 1 \rightarrow A = -\frac{1}{2}$$

$$C = 1$$

$$B = \frac{3}{2}$$

$$X(z) = \underbrace{-\frac{1}{2}}_{a^k} + \underbrace{\frac{\frac{3}{2}z}{z+2}}_{\frac{3}{2} \cdot \frac{z}{z+2}} + \underbrace{\frac{z}{z-1}}_{\varepsilon[k]}$$

$$\underline{\underline{f_n = -\frac{1}{2} \varepsilon[k] + \frac{3}{2} \cdot (-2)^k \varepsilon[k] + \varepsilon[k]}}$$